

IV	Life insurance and annuity contracts		10	15
	21	Continuous Life insurance contracts		
	22	Term life assurance, Endowment		
	23	Whole life, Continuous Life annuities- whole lie annuity		
	24	n-year temporary life annuity,		
	25	n- year certain and life annuity		
	Sections from References:			
V	Open Ended Module:		12	
	1			
	Sections from References:			

Books and References:

Textbook:

1. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, Universities Press.

Reference:

1. Rotar, V.I. (2015). Actuarial Models – The mathematics of Insurance – Second Edition. CRC Press, New York.
2. Promislow, S.D. (2015). Fundamentals of Actuarial Mathematics- Third Edition. John Wiley & Sons, New York.
3. Bowers, N.L., Gerber, H.U., Hickman, J.C., Jones, D.A.& Nesbitt, C.J. (1997). Actuarial Mathematics, Society of Actuaries.

SEMESTER VIII

Programme	B. Sc. STATISTICS				
Course Code	STA8 EJ 411				
Course Title	STATISTICAL METHODS FOR MACHINE LEARNING				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic knowledge in Statistics and programming skills in Python				
Course Summary	Understanding Machine learning using Statistics				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Describe the application of statistical concepts and methods in machine learning	U	F	Seminar Presentation /
2.	Apply various popular machine learning methods to practical situations	Ap	C	Instructor-created exams
3.	Apply the model assessment methods in machine learning techniques	Ap	P	Group Tutorial Work
4.	Connect computing software into machine learning problems	An	P	Home Assignments
5.	Explain basic concepts of Neural Networks in machine learning	U	F	Practical Skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Statistical Learning		10	
	1	Variable types; Predictors, Features, Responses, Quantitative variables, Categorical variables, Ordered categorical variables		
	2	Approaches to prediction; Least squares and nearest neighbors		
	3	Supervised and Unsupervised learning		
	4	Regression and classification problems		
	5	Assessing model accuracy, Mean square error, The bias-variance trade off		
	6	Comparison of linear regression with K-Nearest Neighbors(KNN) regression		
II	Classifications		14	
	7	Classification; concepts and its appropriateness in the case of qualitative responses		

	8	Th logistic model		
	9	Linear Discriminant Analysis (LDA) with only one predictor		
	10	Confusion matrix		
	11	Comparison of logistic regression and LDA methods		
	12	Cross validation; Leave-one-out cross validation, K-Fold cross validation		
	13	Decision trees, Regression trees, Classification trees		
	14	Bagging, Random Forests, Boosting.		
III	Support Vector Machines and Clustering		10	
	15	Maximal margin classifier		
	16	Support vector classifier		
	17	Support vector machines		
	18	K-means clustering		
	19	Hierarchical clustering		
IV	Neural Networks		14	
	20	Neural Networks; The Basic Architecture of Neural networks		
	21	The perceptron, Activation and Loss functions		
	22	Multi-Layer Neural Networks		
V	Open –Ended		12	
	Apply machine learning to real-life projects using software packages in R or Python. (Based on reference books)			

Text Book

- Hastie, T., Tibshirani, R. and Friedman, J. (2017).**
The Elements of Statistical Learning: Data Mining, Inference and Prediction, 2nd edition. Springer, New York
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013).** An Introduction to Statistical Learning with Applications in R. Springer, New York
- Charu C. Aggarwal (2018).** Neural Networks and Deep Learning: A Textbook, Springer

Reference

- Burger, S. V. (2018).** Introduction to Machine Learning with R, O'Reilly Media, Inc.
- Avila, J., Hauck, T. (2017).** Scikit-learn Cookbook: Over 80 Recipes for Machine Learning in Python. Packt Publishing, UK